

Q1 (2023 T1 Final Exam Q4a)

A coupling capacitor is used to block DC current from an amplifier to a speaker as shown in Figure 1. If $V_s = 9\angle 45^\circ \text{ V}$, what is the average power delivered to the speaker?

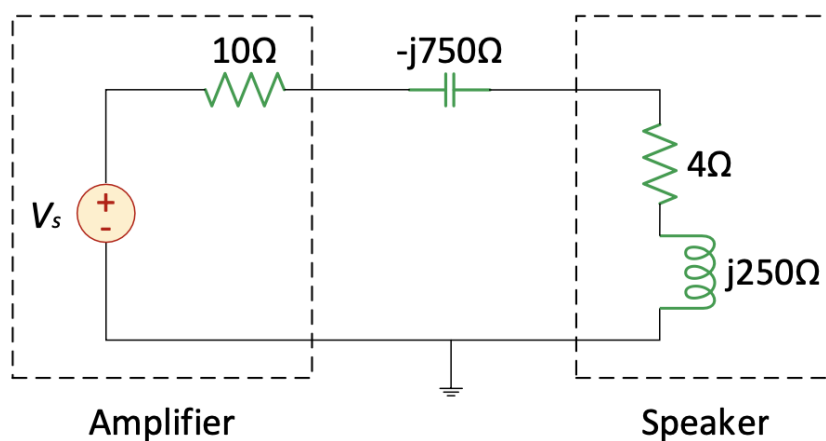


Figure 1

Answer: $P_{\text{speaker}} = 648 \mu\text{W}$

Q2 (2021 T1 Final Exam Q5b)

Given the circuit in Figure 2, find the average power absorbed by the 10-ohm resistor.

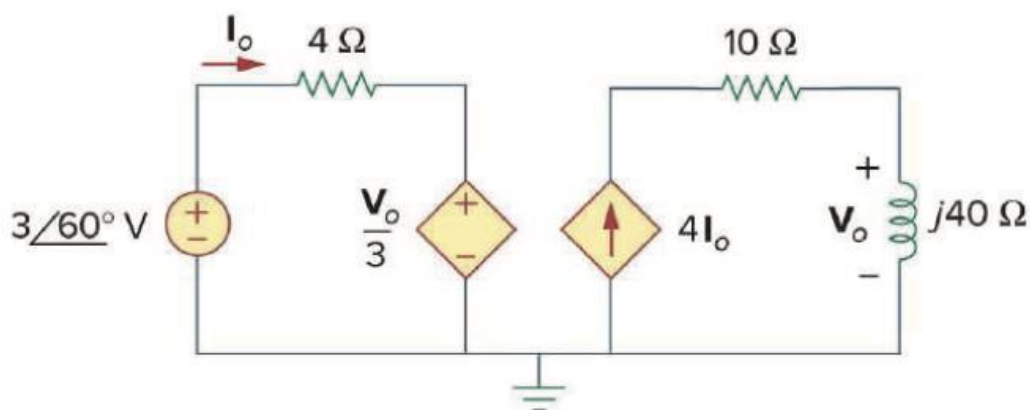


Figure 2

Answer: $P_{10\text{-ohm}} = 0.251 \text{ W}$

Q3 (2021 T1 Final Exam Q5a)

A coupling capacitor is used to block the DC current from an amplifier as shown in Figure 3(a). The amplifier and the capacitor act as the source, while the speaker is the load as in Figure 3(b).

i) Calculate the value of R_L and the source frequency such that the circuit can deliver maximum average power to the speaker.

ii) If $|V_{in}| = 4.6 V_{rms}$, how much power is delivered to the speaker at that frequency?

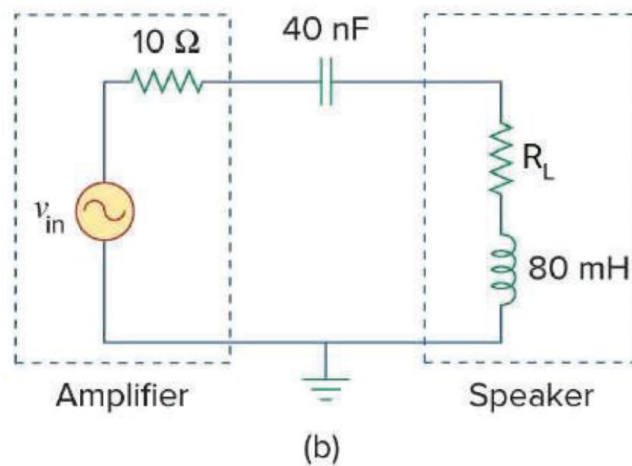
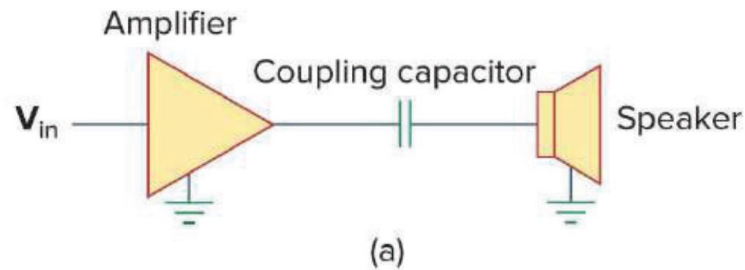


Figure 3

Answers: i) $R_L = 10 \Omega$, $\omega = 17677.67 \text{ rad/s}$ ii) 0.529 W

Q4 (2021 T1 Final Exam Q3b)

Calculate the output voltage $v_o(t)$ in the circuit shown in Figure 4.

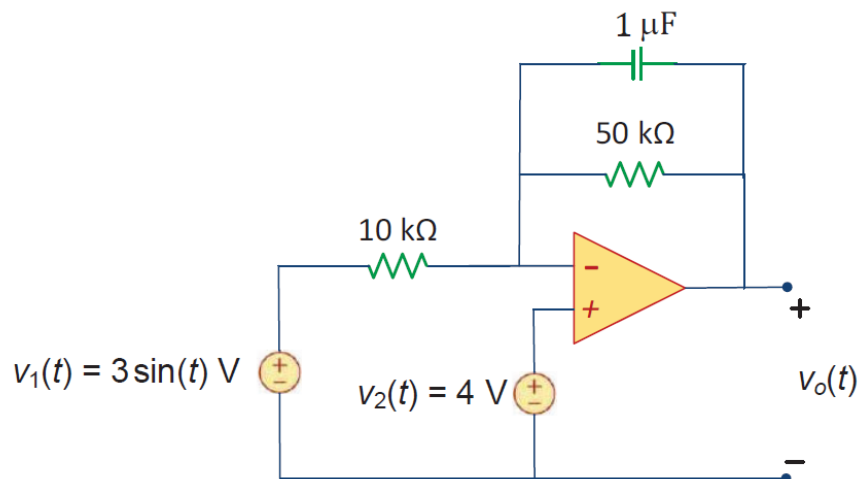


Figure 4

Answer: $v_o(t) = 24 + 14.98\cos(t + 87.14^\circ)\text{ V}$

Q5 (2019 T1 Final Exam Q3b)

Calculate V_o in the circuit shown in Figure 5.

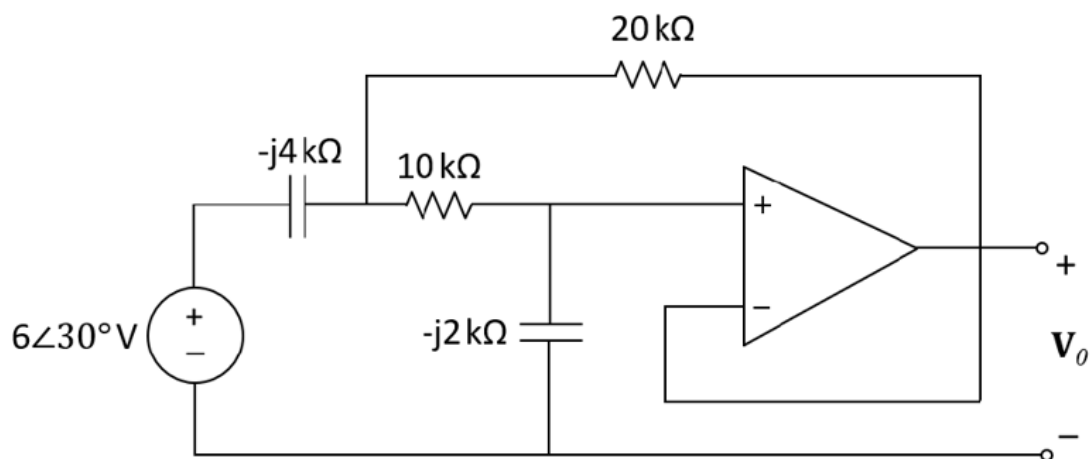


Figure 5

Answer: $V_o = 0.934\angle -21.38^\circ\text{ V}$